Ramakrishna Mission Vivekananda Educational and Research Institute

Big Data Analytics

Batch of 2022 (MSc) Students' Profile

Courses :

Upto Second Semester:

- Machine Learning
- Computer Vision
- Optimisation Algorithms
- Inferential Statistics
- Multivariate Statistics
- Enabling Technologies for Big Data Computation (MySQL, PySpark, Neo4j)
- Supply Chain Management
- Optimisation for Machine Learning
- Linear Algebra
- Data Structures and Algorithms
- Programming for Data Science (R, Python)
- Probability and Stochastic Process

Common Projects :

Upto Second Semester

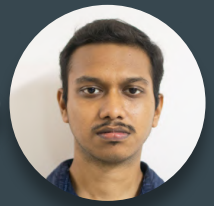
- Univariate and Multivariate Regression using SciKit-Learn
- Classification of MNIST and Fashion MNIST datasets using Multilayer Perceptron
- Image Size Reduction using PCA
- Twitter Sentiment Analysis
- Camera Calibration and Fundamental Matrix Estimation with RANSAC
- Detecting Harris Corner and Matching Images
- Detecting feature points using SIFT (Scale Invariant Feature Transform)
- Image filtering and Hybrid Images

Batch of 2022

Alimpan Barik

Aritra Raut

Dibyendu Das



Dipan banik

Dristanta Das

Mahendra Nandi

Nilabjanayan Bera

Nilotpal Sarkar

Oishik Dasgupta

Pratik Karmakar

Sawan Aich

Sourav Bhattacharjee

Sourav Karmakar

Sourjya Chatterjee

Srijan Mallick

Reach the placement cell at: bdaplacement@gm.rkmvu.ac.in

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Resume

Alimpan Barik

Email: barikalimpan@gmail.com
GitHub: alimpan
LinkedIn: alimpan-barik-b23392184/

Projects:

1. Study of change in out-of-pocket healthcare expenditure for inpatient and outpatient services among households in the Delhi NCR region [Supervisor: Dr. Santanu Pramanik, NCAER] (August 2021 - Present)
2. Segmentation of Brain Tumours from 3-d MRI scans using Deep Learning (March 2021-June 2021) (Here)
3. Comparison between different clustering techniques (K-Means and Gaussian Mixture Models/ EM algorithm) and an approach to develop a new clustering technique
4. EDA of Air quality of major Indian Cities in the period of 2015-2019 using R (November 2020)
5. EDA on worldwide suicide rate dataset in R and Python (October 2020)

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MySQL, Neo4j, PySpark,ML,DL,C,Scikit-Learn, Optimization algorithms

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA-9.67)
BSc(H) Mathematics (Jadavpur University, 2020) (CGPA-9.2)

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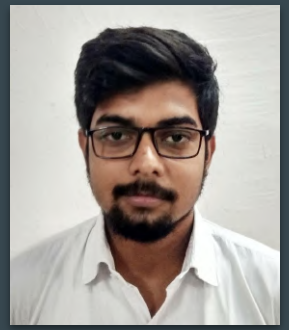

Resume

Aritra Raut

Email: aritraraut@gmail.com
GitHub: aritraraut
LinkedIn: aritra-raut-b770a1190/

Projects:

1. Low back pain analysis using machine learning techniques [Supervisor : Dr. Utpal Garain, CVPR Unit, ISI Kolkata] (August 2021 - Present)
2. Vision Powered book cataloguing system (using YOLOv1 and PyTesseract OCR) [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata] (April 2021 - July 2021) (Here)
3. Factor analysis on Helsinki Running Event dataset (May 2021)
4. Olympic data visualisation in R (November 2020)
5. EDA on Play store app data in R and Python (October 2020)
6. Image Steganography using integer wavelet transformation (November 2018 - April 2019)
7. Online Shopping system using HTML, JavaScript, MySQL.

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MySQL, Neo4j, PySpark, Java, C, HTML, JavaScript, CSS

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA- 8.33)
BSc Computer Science (RKMRC, 2020) (CGPA- 8.54)

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Resume

Dibyendu Das

Email: dsdibyendu7@gmail.com
GitHub: Dibyendu1997
LinkedIn: dibyendu-das-9446a79a/

Projects:

1. Scene Text Detection and Recognition [Supervisor : Dr. Ujjwal Bhattacharya, CVPR Unit, ISI Kolkata] (August 2021 - Present)
2. Application of Barlow Twins to train an encoder using Self Supervised Learning (July 2021 - Present)
3. Bengali Digit Recognition : A Hidden Markov Model Approach (March 2021 - June 2021) (Here)
4. A Simple Pooling-Based Design for Real-Time Salient Object Detection(March 2021 - June 2021)
5. EDA on Covid-19 Dataset (October 2020)
6. A Study on Probabilistic Machine (Jan 2019 - May 2019)

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV,, MySQL,Neo4j, PySpark, NLTK, HmmLearn)

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA-8.67)
MSc Mathematics (Visva Bharati , 2020) (CGPA - 8.88)
BSc Mathematics (Acharya Prafulla Chandra College , 2018) (80%)

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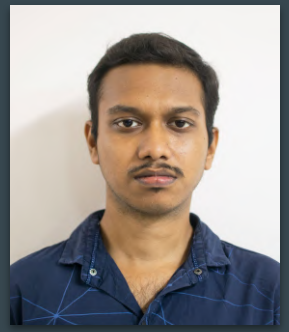

Resume

Dipan Banik

Email: dipanbanik104@gmail.com
GitHub: Dion11235
LinkedIn: dipan-banik

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV, Numba, SimpleITK), MySQL, Neo4j, PySpark, MongoDB

Projects:

1. Improving Arrhythmia prediction on Echocardiograms [Supervisor : Dr. Ujjwal Bhattacharya, CVPR Unit, ISI Kolkata] (August 2021 - present)
2. Left Ventricle Segmentation using fundamental theorem of Neural Network and dynamic programming [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata] (April 2021 - July 2021) (Here)
3. A brief analysis on Simpson's Paradox and how to deal with it (May 2021)
4. Sea-level Pressure Prediction in Austin using Univariate and Multivariate Linear Regression with scikit-Learn (August 2021) (Here)
5. Classification of Bankruptcy Dataset using LDA in R-studio (May 2021)
6. Exploratory data analysis on cardiovascular diseases dataset in R and Python (October 2020)
7. Detailed study on the effect of El-Nino in the climate of Australia in past 10 years (October 2020)

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 9.00)
BSc Mathematics (Asutosh College, Calcutta University, 2020) (62%)

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Resume

Dristanta Das

Email: dasdristanta13@gmail.com
GitHub: dasdristanta13
LinkedIn: dasdristanta13/

Projects:

1. Covid detection using Cough Sound [Supervisor : Dr. Cheruvu Siva Kumar, Department of Mechanical Engineering, IITKGP] (August 2021--Present)
2. 2.5D Visual Sound [Mono to Binaural Audio Conversion] [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata] (February 2021-- June 2021) (Here)
3. Brain Tumour Classification from Brain MRI (August 2021--Present)
4. Optimisation Algorithm Insights (August 2021) (Here)
5. Classification of Bankruptcy Dataset using LDA in R-Studio. (May 2021)
6. EDA on Zomato Restaurants Data (September 2020)
7. EDA on Player Data of FIFA21 Game (November 2020)

Skills:

R (ggplot,dplyr), Python (NumPy, Pandas, Matplotlib, Seaborn, OpenCV, SciKit-Learn, PyTorch,Tensorflow, Keras), MySQL, Neo4j, PySpark,CVAT

Background:

M.Sc. Big Data Analytics (RKMVERI , 2022) (SGPA - 7.67)
B.Sc. Mathematics (Hons.) (Presidency University, 2020) (CGPA - 7.05)

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Resume

Mahendra Nandi

Email: mahendranandi.rkma@gmail.com
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LinkedIn: mahendra-nandi-7038b8176/

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MySQL, Neo4j, PySpark

Projects:

1. CitationIE: Leveraging the Citation Graph for Scientific Information Extraction [Supervisor: Dr. Debarshi Kumar Sanyal, Assistant Professor, School of Mathematical & Computational Sciences, IACS] (Aug 2021 - Present)
2. Image Captioning Using Deep Learning (ResNet50, LSTM) (April 2021 - July 2021) (Here)
3. Classification on Iris dataset (using LDA and QDA) in R (May 2021) (Here)
4. Red Wine Quality (Simple and clean practice dataset for regression or classification modelling) data visualisation in R (November 2020) (Here)
5. EDA and data visualisation in R on **Goodreads-books** (comprehensive list of books listed in goodreads) (October 2020) (Here)

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 7.33)
BSc Physics (Krishnath College, University Of Kalyani, 2019) (74.38%)

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Resume

Nilabjanayan Bera

Email: nilabjanayanbera@gmail.com
GitHub: Nilabjanayan
LinkedIn: nilabjanayan-bera-b74347202/

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MySQL, Neo4j, PySpark, C, HTML

Projects:

1. Summer Project : Machine Learning assisted examination of peripheral blood smear images [Supervisor : Dr. Utpal Garain, CVPR Unit, ISI Kolkata] (August 2021 - Present)
2. Medical image segmentation with EDPCNN (A combination of U-Net and Dynamic Programming in active contour approach) (April 2021 - July 2021) [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata](Here)
3. Salient object detection using Poolnet (June 2021)
4. A simulation study on Non-Parametric test statistics (April 2020)(Here)
5. Descriptive statistical analysis on WHO Life Expectancy data (November 2020)
6. Factor analysis on Helsinki Running Event dataset (May 2021)
7. "Make your own Vector Map" with the help of two way ANOVA model (June 2021)(Here)

Background:

MSc Big Data Analytics (RKMVERI,Belur, 2022) (SGPA - 9.33)
BSc Statistics (RKMRC,Narendrapur, 2020) (CGPA - 7.84)

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Resume

Nilotpal Sarkar

Email: versatilenilot@gmail.com
GitHub: Nilotpal1998
LinkedIn: nilotpal-sarkar-b27085192/

Projects:

1. Information Extraction using Pre-Trained Transformers [Supervisor: Dr. Debarshi Kumar Sanyal, Assistant Professor, School of Mathematical & Computational Sciences, IACS] (Aug 2021 - Present)
2. Regression in Deep Learning Approach (August 2021 - Present)
3. Recommendation Engine on MovieLens Dataset (using Keras and TensorFlow) (April 2021 - July 2021) (Here)
4. Covid-19 detection from Lung Radiograms (using ResNet) (July 2021)
5. Whatsapp Chat Analysis (April 2021) (Here)
6. Template Matching (OpenCV Libraries) (March 2021) (Here)
7. Suicide Rate data Visualisation and Analysis in R and Python (November 2020)
8. EDA on Indian Agricultural Data using R and Python (October 2020)
9. Inventory Management Analysis (March 2018)

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MySQL, Neo4j, PySpark, C, HTML, Macro Economics, Micro Economics, Trade, Econometrics

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 8.00)
BSc Economics (RKM Vidyamandira, 2020) (CGPA - 8.91)

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Resume

Oishik Dasgupta

Email: oishik.dasgupta97@gmail.com
GitHub: oishik25
LinkedIn: oishikdasguptaneil25/

Projects:

1. Digital Topology and Geometry and its application in Biomedical Image Analysis [Supervisor: Prof. Subhadip Basu, Department of Computer Science Engineering, Jadavpur University] (August 2021 - Present)
2. 2D Hand Pose Estimation using U-Net like architecture (April 2021 - July 2021) (Here)
3. Factor analysis on Helsinki 2005 Olympics Running Event dataset (May 2021)
4. Brief analysis of Simpson's Paradox in Covid Data Analysis of Italy and China and how to deal with it. (May 2021)
5. Analysis of IBM HR Attrition Data in R (EDA) (November 2020)
6. Analysis of Cardiovascular Dataset using R and Python (EDA) (October 2020)
7. Detection of Bradycardia and Tachycardia using low-cost Heartbeat Monitor using Arduino UNO (November 2018 - April 2019) [Supervisor: Dr. Suman Halder, Department of Electrical Engineering, NIT Durgapur]

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), Excel, Powerpoint, MySQL, Neo4j, PySpark, C, C++, HTML, CSS, LATEX, Machine Learning, Deep Learning

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 8.33)
BTech Electrical Engineering (NIT Durgapur, 2019) (CGPA - 7.45)

- Automotive Software Engineer Trainee at KPIT (July 2019 - January 2020)

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Resume

Pratik Karmakar

Email: pkpratikkarmakar@gmail.com
GitHub: pratik2358
LinkedIn: pk-pratik-karmakar

Projects:

1. Designing and understanding fundamental limits of online privacy attack [Supervisor: Prof. Debabrota Basu, Faculty, INRIA, France] (August 2021 - Present)
2. Application of Barlow Twins to train an encoder for natural language classification task using Self Supervised Learning (July 2021 - Present)
3. Optimisation Algorithms Insights (August 2021) (Here)
4. Text Generation From Knowledge Graph using Graph Transformers (April 2021 - July 2021) (Here)
5. A lucid visual walk through Fourier Transform (April 2021) (Here)
6. Classification of Cereals (using LDA) in R (May 2021)
7. EDA on worldwide Life Expectancy and Healthcare dataset (by WHO) in R (November 2020)
8. EDA on worldwide suicide rate dataset in R and Python (October 2020)

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 8.67)
BE Electrical Engineering (Jadavpur University, 2018) (CGPA - 8.58)

- Asst. Manager at Vedanta Alumina Limited (2018-2019)
- Summer research intern at DRDO (2017)

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV, nlpaug, NLTK, TorchText), Machine Learning, Deep Learning, MySQL, Neo4j, PySpark, LATEX

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Resume

Sawan Aich

Email: sawanaich@gmail.com
GitHub: Sawan278
LinkedIn: sawan-aich-bda/

Projects:

1. A Novel Gene Embedding Model for Single Cell Transcriptome Analysis [Supervisor: Prof. Dibyendu B. Seal, Assistant Professor, AKCSIT, University of Calcutta](August 2021 - Present)
2. Topic Modelling on Whatsapp Chats using LDA (August 2021 - Present)
3. Easy Visual Question Answering and its Implementation using Flask API [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata](March 2021 - July 2021) (Here)
4. Formation of Hybrid Images (February 2021)
5. EDA on Weather Conditions of Boston, Massachusetts in R (November 2020)
6. Gender Recognition through Voice using Python (August 2019 - April 2020)

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, Pytorch, Keras, SciKit-Learn, OpenCV, Gensim, NLTK), Java, Neo4j, MySQL, PySpark, DBMS, MS-Office (Excel, Powerpoint, Word)

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 8.33)
BSc Computer Science (St. Xavier's College, Kolkata (Autonomous), 2020) (CGPA - 8.26)

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Resume

Sourav Bhattacharjee

Email: sourav11011998@gmail.com
GitHub: Sourav-B98
LinkedIn: sourav-bhattacharjee-6a57811ba/

Projects:

1. 3D Object Reconstruction From 2D Multi-View Images Using GAN [Supervisor : Prof. Bhabatosh Chanda, CVPR Unit, ISI Kolkata] (August 2021 - Present)
2. 3D MRI Image Analysis For Alzheimer's Disease Prediction [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata] (April 2021 - July 2021) (Here)
3. Classification using LDA(Linear Discriminant Analysis) and Various Other Data Engineering on Iris Dataset in R (July 2021) (Here)
4. Creating a Vector Map of University Campus Using Two Way ANOVA Model (July 2021) (Here)
5. Exploratory Data Analysis and Visualisation on Covid 19 Dataset in European Countries using R. (October 2020) (Here)
6. Data Visualisation and Predicting Minimum Cost of an Air Flight from Any Given Location and Destination Using R and Python. (October 2020)

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MS Excel, MS Powerpoint, MySQL, Neo4j, PySpark, Machine learning, Deep Learning.

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 8.33)
BSc Mathematics (Hons.) (Scottish Church College, University of Calcutta, 2020) (79.37%)

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Resume

Sourav Karmakar

Email: sourakarmakar@gmail.com
GitHub: Sourav89068
LinkedIn: sourav-karmakar-693b1a1bb/

Projects:

1. Automatic speech recognition online and offline API integration for developers [Supervisor: Mr. Purnendu Mukherjee of NVIDIA, Santa Clara, California, United States] (August 2021 - Present)
2. Facial Keypoint Detection using CNN [deployed locally with Flask] [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata] (April 2021 - Jul 2021) (Here)
3. Bankruptcy-classification(March 2021 - April 2021) (Here)
4. Computer Vision Projects(February 2021 - May 2021) (Here)
5. EDA on Zomato Restaurant data in R and Python (September 2021 - December 2021) (Here)
6. EDA on Bank Loan Acceptance in R (September 2021 - December 2021) (Here)

Skills:

R (ggplot,dplyr), Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MySQL, Neo4j, Hadoop, PySpark, Java, C, HTML, JavaScript, Flask, LATEX, Sheets, Slides

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA - 8.00)
BSc Mathematics (Banwarilal Bhalotia College, 2020) (CGPA - 7.83)

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Resume

Sourjya Chatterjee

Email: sourjyachatterjee1998@gmail.com
GitHub: SourjyaChatterjee
LinkedIn: sourjya-chatterjee-48174318b

Projects:

1. Logo Synthesis and Manipulation with GAN [Supervisor: Br. Tamal, Department of Computer Science, RKMVERI] (August 2021 - Present)
2. End to End Cell Counting Architecture for Microscopic Image Analysis [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata] (April 2021 - July 2021) (Here)
3. Factor Analysis on Helsinki 2005 Olympics Running Event (Men) DataSet in R (May 2021) (Here)
4. Heart Disease UCI data visualisation in R (November 2020)
5. EDA on Play store app data in R and Python (October 2020) (Here)
6. Optical Character Recognition with CNN (November 2018 - April 2019)

Skills:

R, Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), Machine Learning, Deep Learning, MySQL, Neo4j, PySpark, Java, C, HTML, LateX, Excel, Powerpoint.

Background:

MSc Big Data Analytics (RKMVERI, 2022) (SGPA- 5.67)
BSc Computer Science (RKMRC, 2020) (CGPA- 7.98)

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Resume

Srijan Mallick

Email: srijanmallick98@gmail.com
GitHub: Srijan0098
LinkedIn: srijan-mallick-3508591b2/

Skills:

R (ggplot,dplyr), Python (NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PyTorch, OpenCV), MySQL, Neo4j, PySpark, Minitab, LaTeX

Projects:

1. Multi-label Video Classification [Supervisor : Dr. Ashish Ghosh, Machine Intelligence Unit, ISI Kolkata] (August 2021 - Present)
2. Sentiment Analysis using BERT (August 2021 - Present)
3. Pneumonia Prediction from Chest X-ray Images [Supervisor : Dr. Sujoy Kumar Biswas, Director and Principal Scientist of AIMP Labs, Visiting Scientist at ECSU, ISI Kolkata] (Apr 2021 - July 2021) (Here)
4. Computer Vision mini projects (Mar 2021 - June 2021) (Here)
5. A brief analysis on Simpson's Paradox and how to deal with it (May 2021)
6. Factor analysis on Helsinki Running Event dataset (May 2021)
7. Exploratory data analysis on house prices in R (November 2020)
8. Exploratory data analysis on cardiovascular diseases dataset in R and Python (October 2020)
9. Minimal Cost Network Flow Problems through Graph Theory (Jan 2020 - Apr 2020)

Background:

M.Sc. Big Data Analytics (RKMVERI, 2022) (SGPA - 8.67)
B.Sc. Mathematics (Hons) (SXC, 2020) (CGPA - 7.32)